Density measurement and tritium breeding analysis of ClLiF

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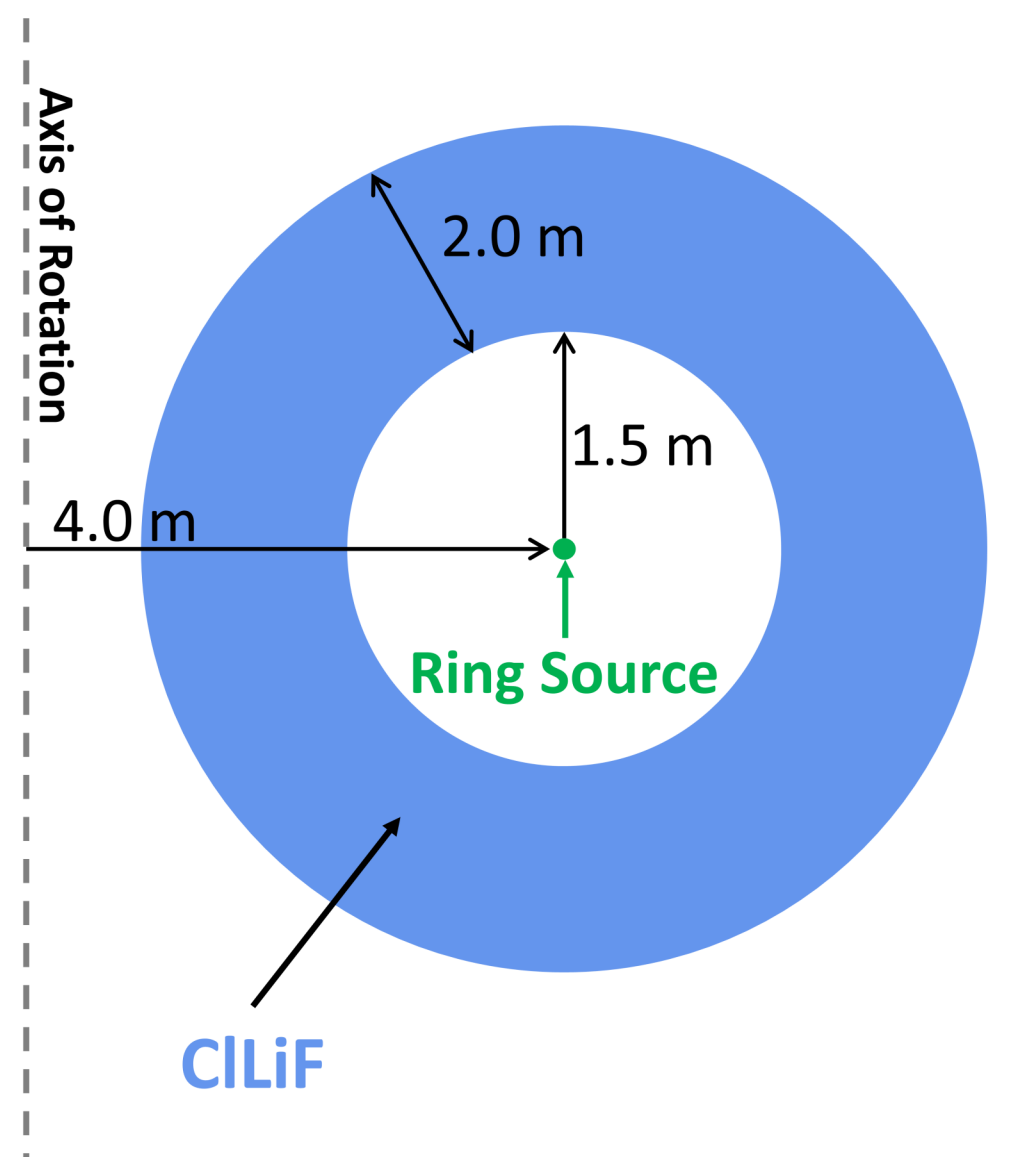
1. ClLiF: Alternative Tritium Breeder

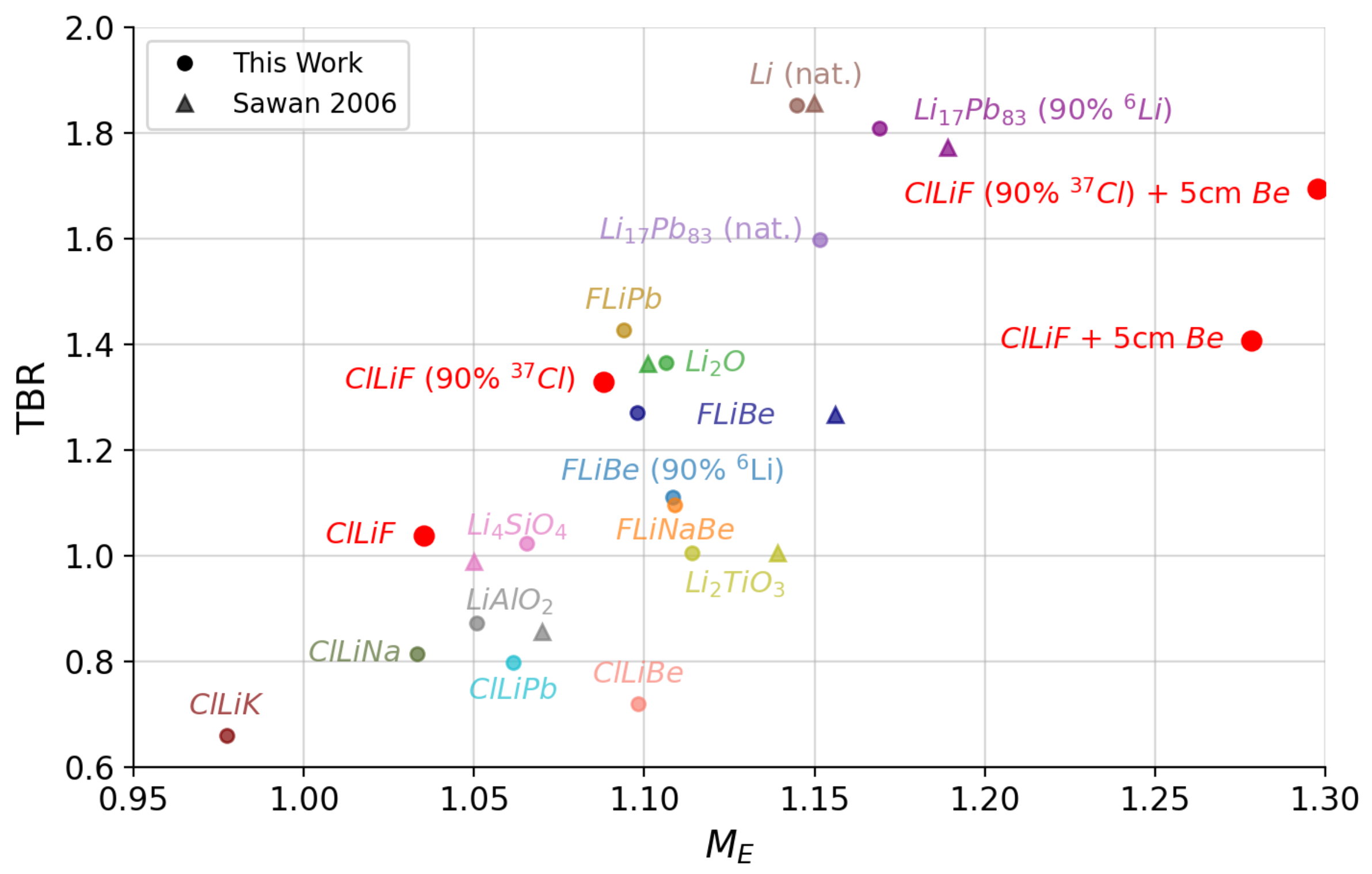
Tritium breeding blankets allow for fusion reactors to produce more tritium fuel than they burn (TBR > 1).

- FLiBe (2LiF + BeF_2), the proposed breeder for ARC (Sorbom 2015), contains highly toxic beryllium

A beryllium-free alternative to FLiBe is ClLiF (LiF + LiCl) molten salt.

- Eutectic at 69.5 at. % LiCl , T_{melt}= 500 °C
- ^{37}Cl acts as a neutron multiplier in DT fusion neutron environment
- In both an ideal blanket model and a more realistic model, **ClLiF enriched in 90% ^{37}Cl performs comparably or better than FLiBe**.





2. Why measure the density of ClLiF

Problem

Only two measurements of ClLiF density in the literature and both measurements lack:

- Direct measurement of eutectic density
- Density from 500-660 °C

Additionally, there is disagreement between these measurements by up to 0.76%.

Solution

Part 1: Computational

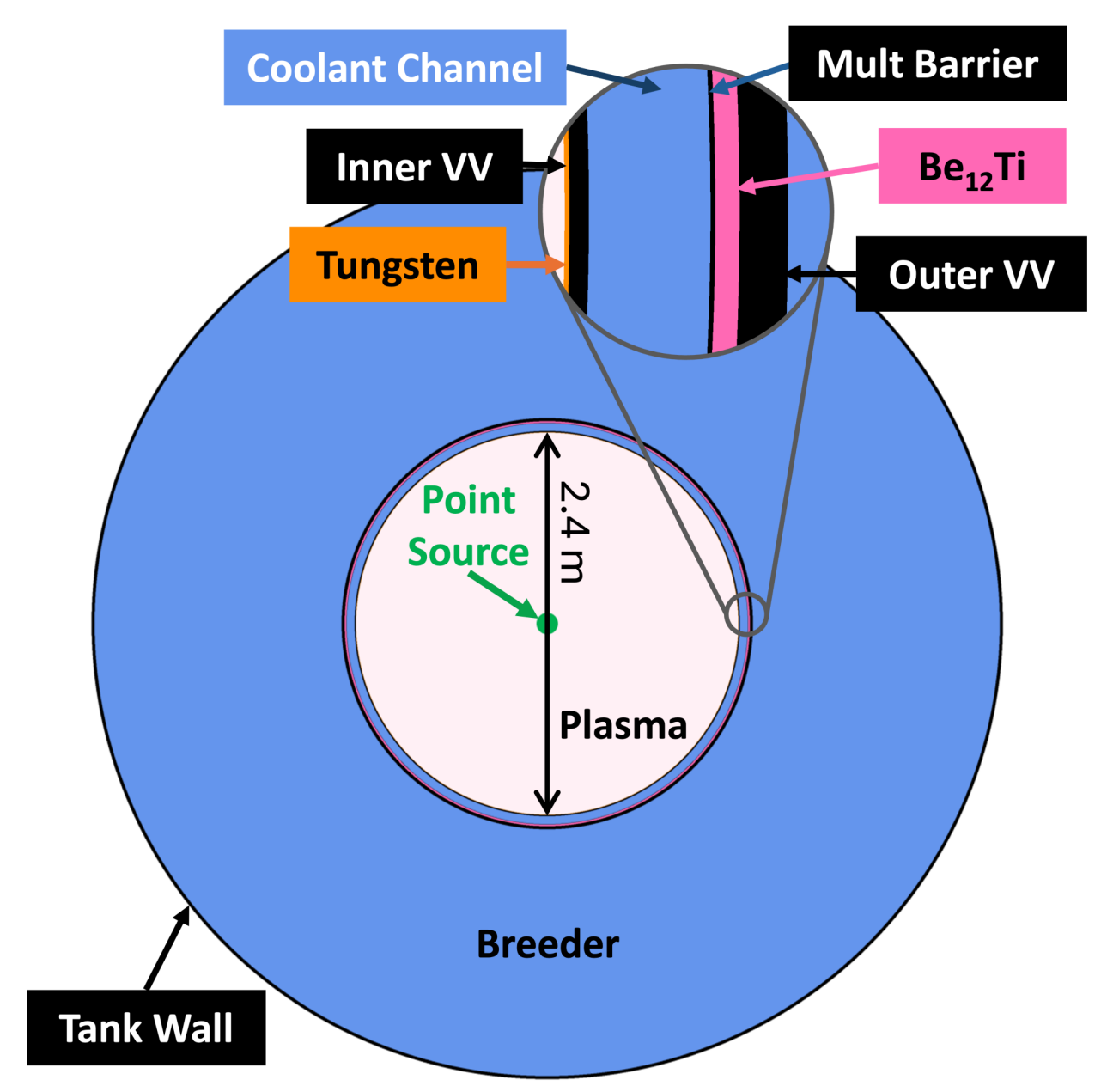
Estimate the impact of density uncertainty on TBR and M_E by using a Total-Monte-Carlo-like technique by running hundreds of simulations with randomly perturbed densities.

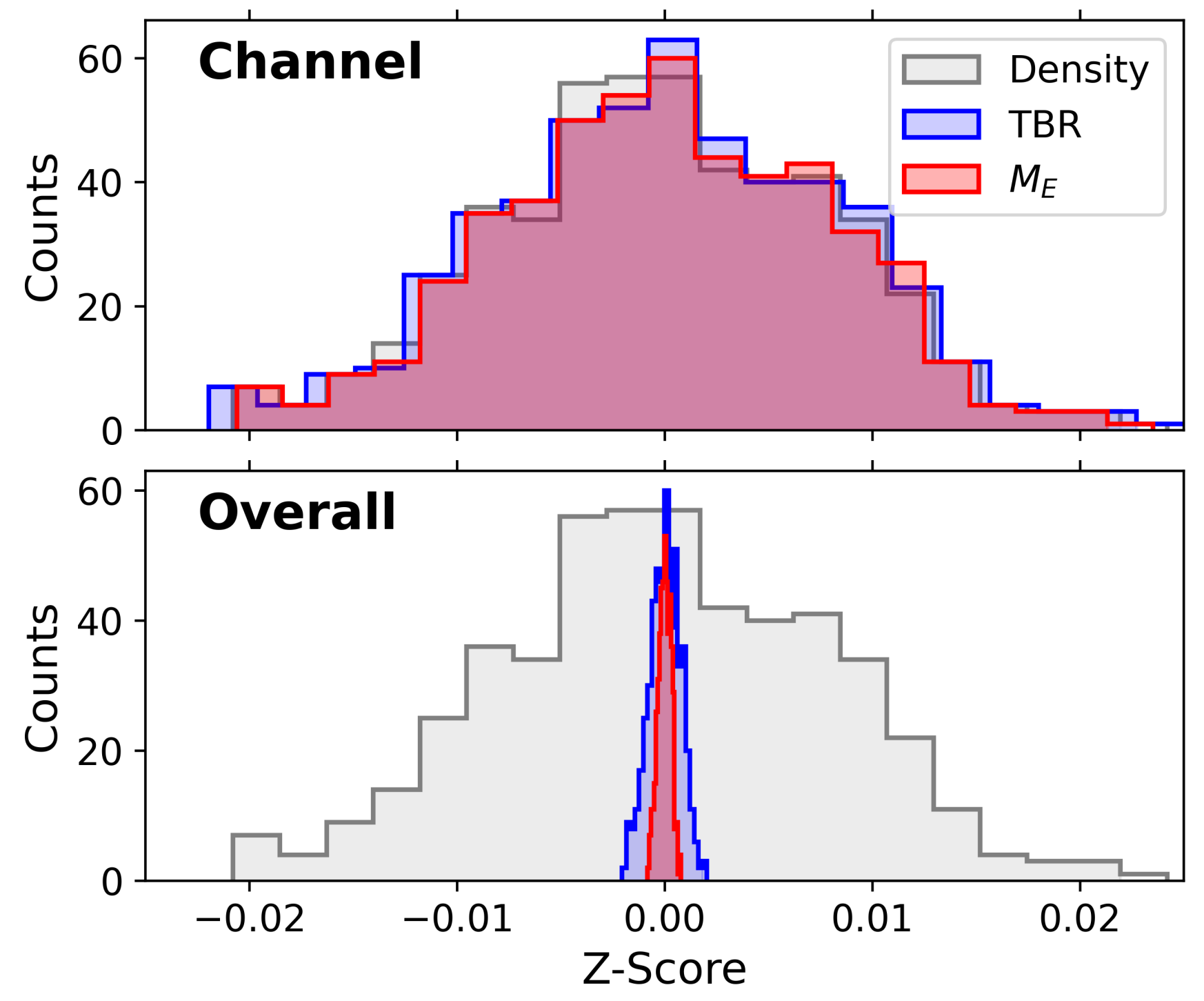
Part 2: Experimental

Measure the density of ClLiF, specifically at eutectic and between 500-660 °C, using Archimedean method, and estimate uncertainty in measurement.

3. Uncertainty Quantification of TBR

- Hundreds of OpenMC simulations were run with perturbed densities to determine effect on TBR and M_E
- Density perturbations randomly sampled from a normal distribution density with a relative standard deviation of 0.76%, the largest deviation between the two ClLiF density measurements in the literature.
- Locally, TBR and M_E uncertainties match with the level of density uncertainty in coolant channel
- In the overall reactor, sensitivities of TBR and M_E to density perturbations are low with 0.07% and 0.03% relative standard deviations

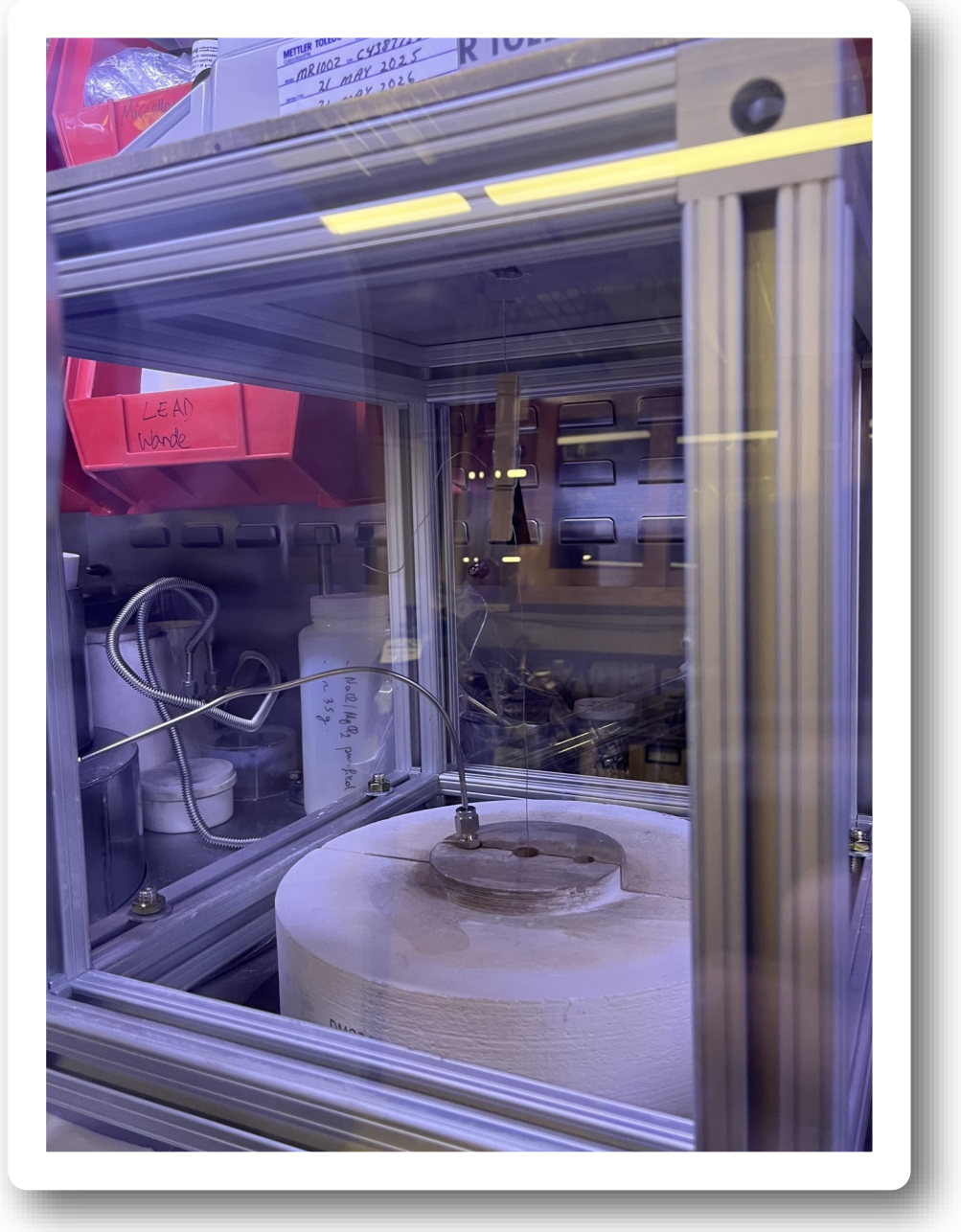


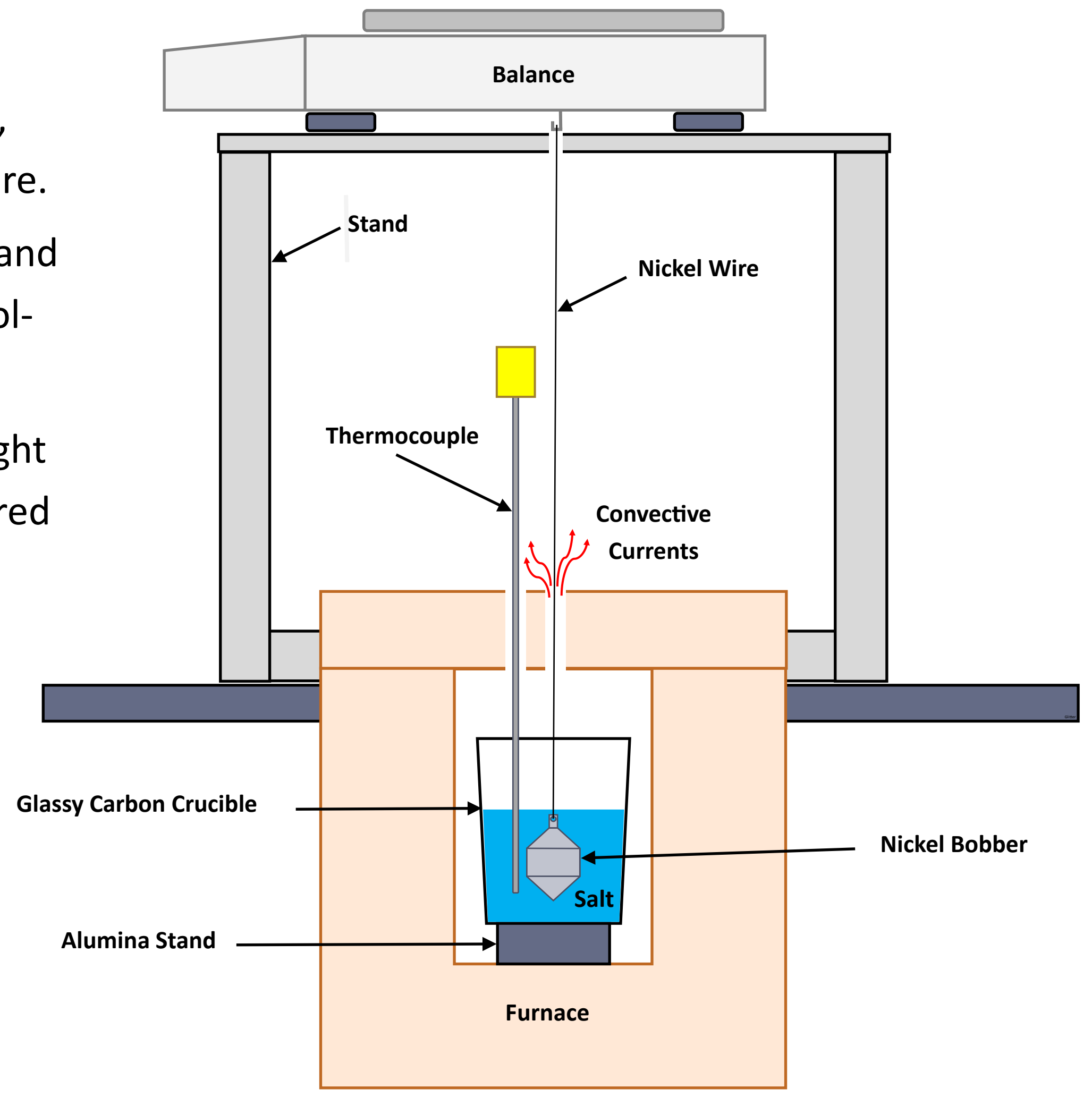


4. Density Experimental Setup

Density was measured through the Archimedean principle based on Vidrio 2022:

1. The weight of a bobber and wire in air, M_{air}, was measured at room temperature.
2. The salt heated up to temperature, T, and the bobber was submerged into the molten salt
3. After equilibrating (>20 minutes), weight of the submerged bobber, M_{sub}, measured 10 times over 10 minutes





$$\rho(T) = \frac{M_{air} - \left(M_{sub} - \frac{\pi d_{wire}\gamma}{g}\right)}{V_{bob,T_0}\left(1 + \alpha(T - T_0)\right)^3}$$

5. Calibration

The density measurement methodology was calibrated and verified by:

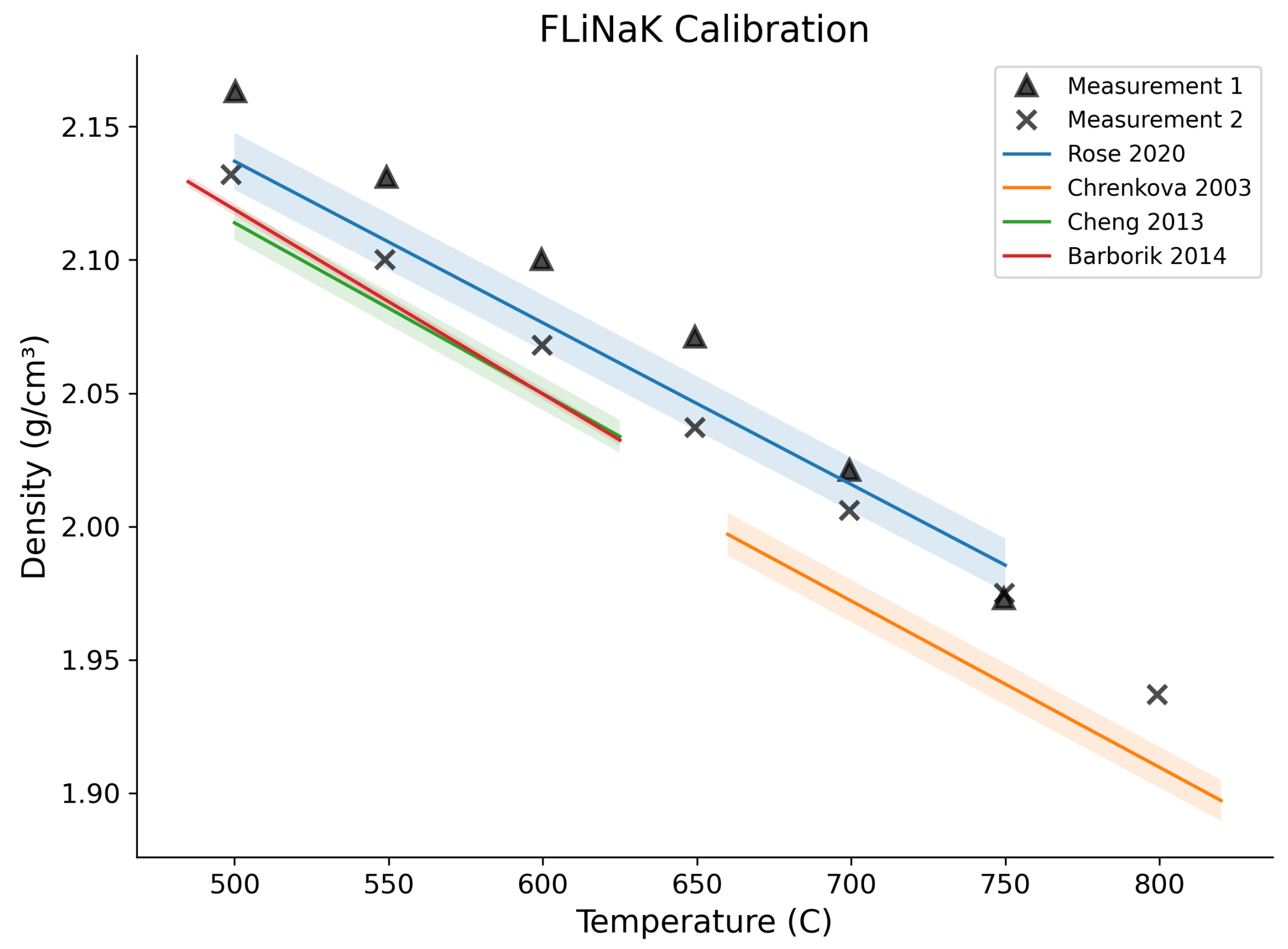
1. In-lab calibration of balance by manufacturer technician with NIST-calibrated weights
2. 200 g NIST calibrated weight hung from balance in glovebox setup was within scale precision
3. Measurement of heavy organic density calibration fluid (2.000 +/- 0.005 g/cc) with experimental setup was 1.994 g/cc at 29 °C.

6. FLiNaK Density Validation

To validate the methodology at high temperatures, the density of FLiNaK (LiF + NaF + KF) salt was measured

Measurement 1: Nickel foil lid placed over salt crucible with holes for the thermocouple and bobber wire. Density values and slope above 650 °C did not agree (9.8×10^{-4} vs. 6.5×10^{-4} g/cc/°C) with literature.

Measurement 2: Foil lid removed. Sediment from furnace in salt. Density values taken during heating matched with literature, but not while cooling down.



The **differing slopes** and **disagreement with literature** and between heating and cooling measurements give **little confidence** in density measurements.

7. Conclusion and Future Work

- Due to lacking literature, ClLiF density should be measured to verify densities used in neutronic analyses
- Local tritium production and heating highly sensitive to density changes near first wall
- Due to failure to validate methodology with FLiNaK, ClLiF density was not measured yet
- Improvements will be made to the experimental setup before repeating validation
  - Make system more rigid (rod instead of wire, holder for crucible)
  - Reduce hot air currents escaping furnace

8. References

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